

Machine Learning-Based Probabilistic Prediction of Tornadoes, Severe Hail, and Severe Thunderstorm Wind Using ERA5-Derived Mesoscale Parameters

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ABSTRACT

Previous studies have shown that severe-weather occurrence is linked to mesoscale thermodynamic, kinematic, and moisture-related conditions. This study evaluates whether ERA5-derived mesoscale parameters can estimate 3-hourly probabilities of tornado, severe hail, and severe thunderstorm wind occurrence across the Ohio River basin. Three-hourly ERA5 reanalysis profiles and gridded fields were used to calculate 107 diagnostic parameters during 1996–2025. NOAA Storm Events reports were paired with ERA5 grid-time diagnostic parameter data using a 25-km radius and forward 3-h window. Random forest (RF) and Extreme Gradient Boosting (XGBoost) models were trained for each hazard using 1996–2018 data, calibrated with 2019–2020 data, and evaluated on 2021–2025 data. In model evaluation, XGBoost outperformed random forest for all hazards based on ROC AUC, average precision, average-precision lift, and Brier Skill Score. XGBoost ROC AUC values were 0.986, 0.987, and 0.978 for tornadoes, severe hail, and severe thunderstorm wind; and average precision values were 0.035, 0.060, and 0.091, respectively. Calibrated probabilities broadly captured the warm-season cycle and west-to-east pattern of observed events. Tornado and severe-hail occurrence rates were highest during March–May, while severe thunderstorm wind occurrence was highest during June–July. Permutation importance ranked 0–3-km shear, 850-hPa moisture transport, and precipitable water as leading tornado predictors; 850–500-hPa lapse rate, precipitable water, and significant hail parameter (SHIP) for severe hail; and precipitable water, SHIP, and 0–6-km lapse rate for severe thunderstorm wind. These results demonstrate the potential for mesoscale diagnostic parameters to support sub-daily probabilistic guidance for severe-weather prediction.